

Math 196 Lecture Notes: Week 7

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0.1 Orthonormal Bases

Recall that a list of vectors $u_1, \dots, u_r$ is said to be *orthonormal* if

$$u_i \cdot u_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

Such vectors are always linearly independent. Indeed, given a linear combination

$$\sum_i x_i u_i = v$$

we have

$$x_j = u_j \cdot v$$

and so if $v = 0$, all $x_j = 0$. If $u_1, \dots, u_r$ span W a subspace, they are called an orthonormal basis for W .

0.2 Orthogonal Projection

Given an orthonormal basis for W , any vector $v \in \mathbb{R}^n$ can be decomposed as $v_1 + v_2$, where

$$v_1 = \sum_i (v \cdot u_i) u_i \in W$$

$$v_2 = v - v_1 \in W^\perp$$

Such a decomposition as an element of W and an element of $W^\perp$ is unique, and we call v_1 the projection of v onto W .

0.3 Graham-Schmidt Process

To compute this in general, we had a recursive process that replaces the vectors in a basis $v_1, \dots, v_r$ for W with new vectors $u_1, \dots, u_r$, one at a time, so that $u_1, \dots, u_r$ is an orthonormal basis for W . Informally, at each step, we use the previously constructed $u_1, \dots, u_i$ to make v_{i+1} orthogonal to all of them, then normalize it. Once we have the orthonormal basis, we can take

$$U = \begin{bmatrix} \vdots & & \vdots \\ u_1 & \dots & u_r \\ \vdots & & \vdots \end{bmatrix}$$

and then projection is given by the matrix $UU^\top$.

1 Algebraic Properties of Transpose

The transpose operation has some algebraic properties that will be useful to us.

1. $(A + B)^\top = A^\top + B^\top$
2. $(cA)^\top = cA^\top$
3. $(A^\top)^\top = A$
4. $(AB)^\top = B^\top A^\top$

The first three of these are straightforward to check from the definition. One can think of (1) and (2) as saying that transpose is linear as an operation on matrices. The fourth one is a bit more interesting. One way to see it is to divide A into rows and B into columns:

$$A = \begin{bmatrix} \dots & r_1 & \dots \\ & \vdots & \\ \dots & r_m & \dots \end{bmatrix}$$

$$B = \begin{bmatrix} \vdots & & \vdots \\ v_1 & \dots & v_n \\ \vdots & & \vdots \end{bmatrix}$$

Then the i, j entry of AB is $r_i v_j$, which is the j, i entry of $(AB)^\top$. On the other hand

$$A^\top = \begin{bmatrix} \vdots & & \vdots \\ r_1^\top & \dots & r_m^\top \\ \vdots & & \vdots \end{bmatrix}$$

$$B^\top = \begin{bmatrix} \dots & v_1^\top & \dots \\ & \vdots & \\ \dots & v_n^\top & \dots \end{bmatrix}$$

and so $B^\top A^\top$ has j, i entry $v_j^\top r_i^\top = r_i v_j$.

In particular, for the projection matrix $P = UU^\top$, we have $P^\top = UU^\top = P$. This is to say that P is a *symmetric matrix*, its entries a_{ij} satisfy $a_{ij} = a_{ji}$.

2 Orthogonal Matrices

Suppose that $u_1, \dots, u_n$ is an orthonormal basis for $\mathbb{R}^n$. We can package these into a matrix

$$A = \begin{bmatrix} \vdots & & \vdots \\ u_1 & \dots & u_n \\ \vdots & & \vdots \end{bmatrix}$$

Because we started with a basis, this matrix is invertible. We can say more in this case. We will first show that A preserves the dot product, in the sense that

$$(Av) \cdot (Aw) = v \cdot w$$

for any vectors $v, w \in \mathbb{R}^n$. To see this, let

$$v = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad w = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

Then

$$v \cdot w = \sum_i x_i y_i$$

On the other hand, we have

$$Av = x_1 u_1 + \cdots + x_n u_n$$

$$Aw = y_1 u_1 + \cdots + y_n u_n$$

and so, by the distributive property for dot product

$$(Av) \cdot (Aw) = \sum_{i,j} x_i y_j (u_i \cdot u_j)$$

now $u_i \cdot u_j$ is 0 if $i \neq j$, and 1 if $i = j$, so this comes out to

$$\sum_i x_i y_i = v \cdot w$$

As a consequence of A preserving the dot product, A preserves length, in the sense that $|Av| = |v|$ for any v . Moreover, A preserves orthogonality, in the sense that if v and w are orthogonal, so are Av and Aw .

In fact, any matrix A which preserves length also preserves orthogonality, because it must take the right triangle with sides $v, w, v + w$ to a right triangle with sides $Av, Aw, Av + Aw$. Such a matrix must take the standard basis $e_1, \dots, e_n$ to another orthonormal basis $u_1, \dots, u_n$. These u_i are the columns of A so we have come full circle.

Conclusion. *The following are equivalent, for A an $n \times n$ matrix:*

1. *The columns of A are an orthonormal basis.*
2. *$|Av| = |v|$ for all v .*
3. *$Av \cdot Aw = v \cdot w$ for all v, w .*

Such a matrix is said to be *orthogonal*. In $\mathbb{R}^2$, we have seen that rotation matrices

$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

are orthogonal. Thinking geometrically, reflection across any line through the origin preserves the length of vectors, so these are orthogonal as well. It turns out that the 2×2 orthogonal matrices are exactly the rotations and reflections. In $\mathbb{R}^3$, rotation about any axis is length preserving, so its matrix is orthogonal. The same goes for reflection across any plane through the origin. Again, these are all the orthogonal 3×3 matrices.

Another useful thing about orthogonal matrices is that they turn out to be easy to invert:

Proposition. *If A is orthogonal, then $A^\top = A^{-1}$.*

Proof. For any matrix A with columns $v_1, \dots, v_n$, the i, j entry in $A^\top A$ is $v_i^\top v_j = v_i \cdot v_j$. If A is orthogonal, this is 0 for $i \neq j$ and 1 for $i = j$, so $A^\top A = I_n$. Multiplying on the right by A^{-1} , we get $A^\top = A^{-1}$. $\square$

In fact, the condition that $A^\top = A^{-1}$ is equivalent to the other properties characterizing orthogonal matrices.

Proposition. *If A and B are orthogonal, then AB and A^{-1} are orthogonal.*

Proof. We check that AB and A^{-1} are length preserving:

$$|ABv| = |A(Bv)| = |Bv| = |v|$$

first using the fact that A is length preserving, then that B is length preserving. Similarly,

$$|v| = |AA^{-1}v| = |A^{-1}v|$$

using the fact that A is length preserving. □

3 Least Squares

3.1 Distance and Least Squared Solutions

Given two vectors v and w , the distance between them (between their endpoints) is the length of their difference, $d(v, w) = |v - w|$. With this view of distance, we can characterize orthogonal projection in a different way.

Proposition. *Let W be a subspace of $\mathbb{R}^n$ and $v \in \mathbb{R}^n$ be a vector. Then $\text{proj}_W(v)$ is the vector $w \in W$ which is closest to v , in the sense that it minimizes $d(v, w)$.*

Proof. Let $w = \text{proj}_W(v)$ and $w' \in W$ be another vector. We want to show that

$$d(v, w') \geq d(v, w)$$

Because $v - w \in W^\perp$ and $w' - w \in W$, these vectors are orthogonal, and so we have a right triangle with corners w', w, v . [Draw picture]. The Pythagorean theorem (or alternatively an explicit computation with dot products) then tells us that

$$|w' - v|^2 = |w - v|^2 + |w' - w|^2 \geq |w - v|^2$$

and so

$$|w' - v| \geq |w - v|$$

as required. □

If we present W as $\text{im}(A)$ for some matrix A , then we can view this in the following way. An equation $Ax = v$ may be inconsistent if $v \notin \text{im}(A)$. In this case, we can try to find something which is as close to a solution as possible. One interpretation of this is to find an x so that $|Ax - v|$ is as small as possible. From the above discussion, this will be true provided that $Ax = \text{proj}_W(v)$. As $\text{proj}_W(v) \in \text{im}(A)$, this equation is always solvable.

It is equivalent to say that we are looking for an x such that $|Ax - v|^2$ is as small as possible. This is the sum of squares of the entries of $Ax - v$, so such an x is called a *least squares* solution to the linear system $Ax = v$.

From what we have seen so far, we can find a least squares solution x via the following steps:

1. Find a basis for $\text{im}(A)$, but bringing A to reduced row echelon form and picking out the pivot columns.
2. Apply the Gram-Schmidt process to this basis to find an orthonormal basis.
3. Use this orthonormal basis to compute the projection w of v onto $\text{im}(A)$.
4. Solve the system $Ax = w$.

This would be a quite laborious process, so it is quite fortunate that there is a more direct way to find least-squares solutions.

3.2 The Normal Equation

Let $Ax = v$ be a linear system. We call the equation

$$A^\top Ax = A^\top v$$

the *normal equation* associated with the original linear system.

Proposition. *A vector x is a least squares solution to $Ax = v$ if and only if it is an exact solution to the normal equation.*

Proof. From our original characterization of orthogonal projection, we know that x is a least squares solution if and only if $Ax - v \in \text{im}(A)^\perp$, as this means that $Ax = \text{proj}_{\text{im}(A)}(v)$. On the other hand, $\text{im}(A)^\perp = \ker(A^\top)$, so this holds if and only if

$$0 = A^\top(Ax - v) = A^\top Ax - A^\top v$$

□

Example 1. Consider the linear system

$$\begin{bmatrix} 1 & 0 \\ 1 & -2 \\ 1 & -1 \end{bmatrix} x = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

We apply row operations to the augmented matrix as follows:

$$\left[\begin{array}{cc|c} 1 & 0 & 0 \\ 1 & -2 & 1 \\ 1 & -1 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & -1 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & -1/2 \\ 0 & -1 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & -1/2 \\ 0 & 0 & -1/2 \end{array} \right]$$

From this, we find that the system is inconsistent. We now look for a least squares solution, so we form the normal equation by applying the transpose of the matrix to both sides:

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & -2 \\ 1 & -1 \end{bmatrix} x = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Computing the matrix products, this is

$$\begin{bmatrix} 3 & -3 \\ -3 & 5 \end{bmatrix} x = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Now we can solve the resulting linear system as follows:

$$\left[\begin{array}{cc|c} 3 & -3 & 1 \\ -3 & 5 & -2 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 3 & -3 & 1 \\ 0 & 2 & -1 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & -1 & 1/3 \\ 0 & 1 & -1/2 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & 1/6 \\ 0 & 1 & -1/2 \end{array} \right]$$

This tells us that $x = \begin{bmatrix} -1/6 \\ -1/2 \end{bmatrix}$ is a least squares solution to the original system. The vector

$$\begin{bmatrix} 1 & 0 \\ 1 & -2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -1/6 \\ -1/2 \end{bmatrix} = \begin{bmatrix} -1/6 \\ 5/6 \\ 1/3 \end{bmatrix}$$

is the closest element of the image to $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$. Here the optimal distance squared is

$$\left\| \begin{bmatrix} -1/6 \\ 5/6 \\ 1/3 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\|^2 = 1/36 + 1/36 + 4/36 = 6/36 = 1/6$$

Let's again consider the normal equation, in generality:

$$A^\top Ax = A^\top v$$

Because solving this is equivalent to solving $Ax = \text{proj}_{\text{im}(A)} v$, this system is always consistent. Moreover, if A is injective, then this system always has a unique solution. Indeed:

Proposition. $\ker(A) = \ker(A^\top A)$ for any A . In particular, if $\ker(A) = \{0\}$, then $A^\top A$ is a square matrix with kernel $\{0\}$, so is invertible.

Proof. Clearly if $Ax = 0$, then $A^\top Ax = 0$. On the other hand, if $A^\top Ax = 0$, then $Ax \in \ker(A^\top) = (\text{im}(A))^\perp$, but also $Ax \in \text{im}(A)$, so we must have $Ax = 0$. $\square$

In the case that A is injective, we can write a formula for the unique least squares solution to $Ax = v$. This is the unique exact solution to $A^\top Ax = A^\top v$, which is $x = (A^\top A)^{-1} A^\top v$. This also gives us a new formula for projection. Recalling that the least squares solution satisfies $Ax = \text{proj}_{\text{im}(A)}(v)$, we have

$$\text{proj}_{\text{im}(A)}(v) = A(A^\top A)^{-1} A^\top v$$

This makes sense provided that A is injective. Thus, one way to compute a matrix for projection onto W is to find a (not necessarily orthonormal) basis of W , then put the vectors into a matrix A . This A will be injective and have $\text{im}(A) = W$, so we have

$$\text{proj}_W = A(A^\top A)^{-1} A^\top$$

This is a generalization of the formula

$$\text{proj}_v = \frac{1}{|v|^2} vv^\top$$

for projection onto a line.

3.3 Data Fitting

We expect linear systems $Ax = v$ to be inconsistent when they have more equations than unknowns. Such systems are called *overdetermined*. One way that overdetermined systems can arise is in the problem of fitting a function with a small number of parameters to a large data set.

For example, suppose that we have some data points (x_i, y_i) , and we have good reason to believe that y is approximately a linear function of x , of the form $y = ax + b$, where a, b are unknown constants. Then each data point yields a linear equation that should be satisfied by a, b . If we have two data points, then there will be a unique line connecting them and we can find a, b by solving a linear equation.

Example 2. Suppose we have data points $(1, 2)$ and $(3, 3)$. These translate into linear equations

$$\begin{cases} a + b = 2 \\ 3a + b = 3 \end{cases}$$

Solving the system, we obtain

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 3 & 1 & 3 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & -2 & -3 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1 & 3/2 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & 1/2 \\ 0 & 1 & 3/2 \end{array} \right]$$

and so $a = 1/2$, $b = 3/2$, and we predict that $y = \frac{x}{2} + \frac{3}{2}$.

If we have more than two data points, each will contribute an equation and so the system will be overdetermined. If y is not exactly equal to $ax + b$ or is imperfectly measured, the system is likely to be inconsistent as well. In this case, a least squares solution gives us a reasonable “best guess” at the true values of a, b .

Example 3. Suppose we add to our data set the point $(-1, 0)$. Now our system is

$$\begin{cases} a + b = 2 \\ 3a + b = 3 \\ -a + b = 0 \end{cases}$$

Or equivalently,

$$\begin{bmatrix} 1 & 1 \\ 3 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$$

We know this is inconsistent: the first two equations already imply that a and b take the values obtained above, but these do not satisfy the third equation. Looking for a least squares solution, we form the normal equation

$$\begin{bmatrix} 1 & 3 & -1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 & 3 & -1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$$

which comes out to

$$\begin{bmatrix} 11 & 3 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 11 \\ 15 \end{bmatrix}$$

We can solve this as follows:

$$\left[\begin{array}{cc|c} 11 & 3 & 11 \\ 3 & 3 & 15 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 3/11 & 1 \\ 3 & 3 & 15 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 3/11 & 1 \\ 0 & 24/11 & 2 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 3/11 & 1 \\ 0 & 1 & 11/12 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & 3/4 \\ 0 & 1 & 11/12 \end{array} \right]$$

Thus $y = \frac{3}{4}x + \frac{11}{12}$ is the line that best fits these data points, in the sense that the sum of the squares of the errors is as small as possible. [Draw picture]

This method can be applied to other families of functions as well. For example, we might expect that our y is a quadratic function of x , $y = ax^2 + bx + c$. Given a list of data points, we could then set up a system of linear equations, and a least squares solution would give us a quadratic function that minimizes the sum of squares of error over all the data points. For this to work, we just need the output of the function to depend linearly on the *parameters*, here a, b, c , so that when we plug in inputs and outputs, we get linear equations in the parameters. There is also nothing stopping us from doing this with a multi variable function, as long as everything is linear in the parameters. In general, this process is known as *linear regression*.

Example 4. Suppose you are on a distant planet with unknown gravity. You throw a stone upwards, letting it fall back to the ground. Using a camera and meter stick, you get the following data points for the vertical displacement of the stone at different times.

t	1	1.5	2	2.5
y	3.5	5	4	2.5

Our prediction for y is $y = vt - \frac{g}{2}t^2$ where v is the initial upward velocity and g is the acceleration due to gravity, both unknown. Plugging in each data point, we get linear equations that must be satisfied by v and g .

$$\begin{cases} v - 0.5g = 3.5 \\ 1.5v - 1.125g = 5 \\ 2v - 2g = 4 \\ 2.5v - 3.125g = 2.5 \end{cases}$$

Or equivalently

$$\begin{bmatrix} 1 & -0.5 \\ 1.5 & -1.125 \\ 2 & -2 \\ 2.5 & -3.125 \end{bmatrix} \begin{bmatrix} v \\ g \end{bmatrix} = \begin{bmatrix} 3.5 \\ 5 \\ 4 \\ 2.5 \end{bmatrix}$$

The normal equation is

$$\begin{bmatrix} 1 & 1.5 & 2 & 2.5 \\ -0.5 & -1.125 & -2 & -3.125 \end{bmatrix} \begin{bmatrix} 1 & -0.5 \\ 1.5 & -1.125 \\ 2 & -2 \\ 2.5 & -3.125 \end{bmatrix} \begin{bmatrix} v \\ g \end{bmatrix} = \begin{bmatrix} 1 & 1.5 & 2 & 2.5 \\ -0.5 & -1.125 & -2 & -3.125 \end{bmatrix} \begin{bmatrix} 3.5 \\ 5 \\ 4 \\ 2.5 \end{bmatrix}$$

This comes out to approximately

$$\begin{bmatrix} 13.5 & -14 \\ -14 & 15.3 \end{bmatrix} \begin{bmatrix} v \\ g \end{bmatrix} = \begin{bmatrix} 25.3 \\ -23.2 \end{bmatrix}$$

This has solution approximately

$$\begin{bmatrix} v \\ g \end{bmatrix} = \begin{bmatrix} 5.8 \\ 3.8 \end{bmatrix}$$

Thus 3.8 is a good guess for the planets gravitational acceleration.

Remark. The answers that this method gives you are only meaningful if your assumption for the relationship between the variables is well justified. Given n data points (x, y) , I can always exactly fit them by a degree $n - 1$ polynomial, but this may not say anything about any underlying phenomenon.

4 Determinants

We are now beginning the third main part of the course. The starting point here is to learn a criterion for when a square matrix is invertible. This will take the form “ A is invertible if and only if $f \neq 0$ ” where f is a polynomial in the entries of the matrix that we will describe. This kind of criterion can be useful for deciding if a particular matrix is invertible, but it is most useful when we have a family $A(t)$ depending on a parameter t , and we need to decide which of them are invertible. This is exactly how we will use it when we find eigenvalues of the matrix.

4.1 Review of Dimension Two

For 2×2 matrices, you saw such a polynomial on the homework. Recall that

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is invertible if and only if $ad - bc \neq 0$. One way to see this is by explicitly applying row operations to the matrix, and keeping track of what needs to be nonzero in order to get to the identity matrix. In this case, we also had a formula for the inverse:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \text{ if } ad - bc \neq 0$$

Let’s give this quantity a name. We call $ad - bc$ the determinant of A , written $\det(A)$.

4.1.1 Geometry of the Determinant in Dimension Two

We can interpret the quantity $ad - bc$ in terms of the geometry of the column vectors

$$v = \begin{bmatrix} a \\ c \end{bmatrix}, w = \begin{bmatrix} b \\ d \end{bmatrix}$$

In particular, it is the dot product:

$$ad - bc = \begin{bmatrix} a \\ c \end{bmatrix} \cdot \begin{bmatrix} d \\ -c \end{bmatrix}$$

The latter vector is the rotation of w by $-\pi/2$. Using our geometric formula for the dot product, this is $|v||w|\cos(\pi/2 - \theta) = |v||w|\sin(\theta)$, where θ is the (signed) angle from v to w . This can be interpreted as the (signed) area of the parallelogram with sides v and w . [draw picture]. It is positive if the relative orientation of v and w is the same as that of e_1 and e_2 , and negative if the relative orientation is opposite. This parallelogram is how the matrix A transforms the unit square with sides e_1, e_2 , so the determinant tells us about how A modifies area and orientation. In the case that the $\det A = 0$, A sends everything to a line with 0 area.

Example 5. A rotation matrix has the form

$$\begin{bmatrix} \alpha & -\beta \\ \beta & \alpha \end{bmatrix}$$

where $\alpha^2 + \beta^2 = 1$. The determinant of such a matrix is $\alpha^2 + \beta^2 = 1$. A reflection matrix has a similar form:

$$\begin{bmatrix} \alpha & \beta \\ \beta & -\alpha \end{bmatrix}$$

This has determinant $-\alpha^2 - \beta^2 = -1$. Both preserve area, rotations preserve orientation and reflections reverse it.

4.2 The Determinant as a Sum over Permutations

The determinant of an $n \times n$ matrix is a generalization of this quantity attached to 2×2 matrices. As in the 2×2 case, this has the following properties:

1. $\det(A)$ is a polynomial in the entries of A with coefficients ± 1 .
2. A is invertible if and only if $\det(A) \neq 0$.
3. $|\det(A)|$ represents the volume of the image of the n -dimensional unit cube under A , and the sign of $\det(A)$ tells us whether A preserves or reverses orientation (whatever this means).

I will give you a definition of $\det(A)$ of the form specified in (1), and we will see why (2) is true later when we connect this definition to row operations. I will not say much more about (3) in higher dimensions.

Definition. A *permutation* of length n is a list of numbers $(k_1, \dots, k_n)$ such that each number from 1 to n appears exactly once in the list.

Example 6. The two permutations of length 2 are $(1, 2)$ and $(2, 1)$. For length 3, we have

$$(1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 1, 2), (3, 2, 1)$$

In general there are $n!$ permutations of length n .

We will use σ to denote a permutation and $\sigma(i)$ the i th entry in σ . Given a permutation σ , we can pick out one entry in each row of an $n \times n$ matrix, by selecting the $\sigma(i)$ th entry of row i for example:

$$(3, 1, 2) \rightsquigarrow \begin{bmatrix} & & * \\ * & & \\ & * & \end{bmatrix}$$

For our purposes, one can think of permutations as being patterns like this. The terms in the determinant will be products of these matrix entries, in the above example $a_{13}a_{21}a_{32}$. We denote this product a_σ . The determinant then has the form

$$\det(A) = \sum_{\sigma} \pm a_\sigma$$

For example, in 2×2 , the permutations are $(1, 2)$, yielding $a_{11}a_{22}$ and $(2, 1)$, yielding $a_{21}a_{12}$, and the determinant is $a_{11}a_{22} - a_{21}a_{12}$.

We now need to explain where the positive and negative signs comes from. For a permutation σ , we denote by $r(\sigma)$ the number of pairs in σ which are “out of order.” This is the number of choices of $i < j$ where $\sigma(i) > \sigma(j)$. In terms of the matrix, this is the number of pairs that look like this:

$$\begin{bmatrix} & & * \\ & * & \\ * & & \end{bmatrix}$$

The *sign* of σ , $\text{sgn}(\sigma)$ is defined to be $(-1)^{r(\sigma)}$. I can now state the definition of the determinant:

$$\det(A) = \sum_{\sigma} \text{sgn}(\sigma) a_{\sigma}$$

Again returning to the case that $n = 2$, our permutations are $(1, 2)$ and $(2, 1)$. The first has nothing out of order, so $\text{sgn}(1, 2) = 1$. The second has one pair out of order, so $\text{sgn}(2, 1) = -1$. This gives us the expected formula $a_{11}a_{22} - a_{21}a_{12}$.

This definition is good for explaining the otherwise miraculous properties of the determinant, but we will learn other ways to compute it which are much more efficient in many circumstances.

4.2.1 Some Easy Cases

For the identity matrix, we can say that $\det(I_n) = 1$ for all n . The only permutation for which a_{σ} is nonzero is $(1, 2, 3, \dots, n)$. This has sign $+1$, and the corresponding product is the product of the diagonal entries, 1.

More generally, we call a matrix upper triangular if it has zeroes below the diagonal, so looks something like

$$\begin{bmatrix} a_{11} & * & * \\ 0 & a_{22} & * \\ 0 & 0 & a_{33} \end{bmatrix}$$

The determinant of such a matrix is the product of the diagonal entries. Indeed, all permutations except $(1, 2, \dots, n)$ pick out something below the diagonal, and so the corresponding term is 0. The same thing is true for lower triangular matrices.

Example 7.

$$\det \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -2 & 1 & 1 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = 1 \cdot (-2) \cdot (-1) \cdot 1 = 2$$

Finally, the determinant of a 1×1 matrix $[a]$ is just the number a . The only permutation is (1) .

4.3 Dimension Three

Let's see what our definition gives us in the case of 3×3 matrices. Consider a matrix

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

There are 6 permutations, so there will be 6 terms in the determinant. There is a nice visual way to organize these terms and get their signs. Imagine the terms of the matrix extending to the right past the edge, repeating each row:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{matrix} a_{11} & a_{12} & \dots \\ a_{21} & a_{22} & \dots \\ a_{31} & a_{32} & \dots \end{matrix}$$

Now draw 3 lines diagonally down to the right, and 3 up to the right. [Draw picture] Each line picks out 3 terms. The downward lines have sign +1 and the upward lines have sign -1. Putting these together, we get

$$a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12} = \det(A)$$

You can check that these 6 terms correspond to the 6 permutations of length 3, with their corresponding signs.

Example 8. Consider the matrix

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & -1 \\ 1 & 0 & 1 \end{bmatrix}$$

Extending it to the right, we get

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & -1 \\ 1 & 0 & 1 \end{bmatrix} \begin{matrix} 1 & 1 & \dots \\ 0 & 2 & \dots \\ 1 & 0 & \dots \end{matrix}$$

The determinant is then

$$\begin{aligned} & 1 \cdot 2 \cdot 1 + 1 \cdot (-1) \cdot 1 + 1 \cdot 0 \cdot 0 - 1 \cdot 2 \cdot 1 - 0 \cdot (-1) \cdot 1 - 1 \cdot 0 \cdot 1 \\ & = 2 - 1 + 0 - 2 - 0 - 0 = -1 \end{aligned}$$

As this is not zero, we can conclude that the matrix is invertible.

4.4 Cofactor Expansion

For larger matrices, there is a useful recursive method of computing the determinant. We can see an example of this if we look at the formula for the determinant of a 3×3 matrix:

$$\det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} =$$

$$a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}$$

In this expression, we can group terms according to which entry in the first row appears:

$$a_{11}(a_{22}a_{33} - a_{32}a_{23}) - a_{12}(a_{21}a_{33} - a_{31}a_{23}) + a_{13}(a_{21}a_{32} - a_{31}a_{22})$$

The other terms look like 2×2 determinants. In particular, this is

$$a_{11} \det \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix} - a_{12} \det \begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix} + a_{13} \det \begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

These 2×2 matrices can be obtained by removing one row and one column from the original matrix, specifically the row and column corresponding to the coefficient outside the determinant. Let's introduce some notation for this. Given a $n \times n$ matrix A , A_{ij} denote the $(n-1) \times (n-1)$ matrix obtained by removing the i th row and j th column. Then in the 3×3 case we can write

$$\det(A) = a_{11} \det A_{11} - a_{12} \det A_{12} + a_{13} \det A_{13}$$

A similar formula works for matrices of any size:

$$\det A = \sum_j (-1)^{1+j} a_{1j} \det A_{1j}$$

This is called a *cofactor expansion* of the determinant. This is a recursive formula, because it reduces the computation of the determinant of an $n \times n$ matrix to the computation of determinants of $(n-1) \times (n-1)$ matrices. This is specifically the expansion across the first row. One can compute the determinant via a cofactor expansion across any row:

$$\det A = \sum_j (-1)^{i+j} a_{ij} \det A_{ij}$$

The only thing that changes is the signs. One can think of the sign of each term in the cofactor expansion as coming from the following pattern in the matrix:

$$\begin{bmatrix} + & - & + & \dots \\ - & + & - & \\ + & - & + & \\ \vdots & & & \ddots \end{bmatrix}$$

There is also a completely analogous formula that expands across a column as well as a row, using the same pattern of signs.

Example 9. Consider the matrix

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 3 & 1 & 3 & 0 \\ 3 & 2 & 2 & 0 \\ 4 & 0 & 0 & 3 \end{bmatrix}$$

We can expand its determinant as a sum across the bottom row: this is a good choice, because there are only two nonzero terms here:

$$-4 \det \begin{bmatrix} 0 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 2 & 0 \end{bmatrix} + 0 - 0 + 3 \det \begin{bmatrix} 1 & 0 & 1 \\ 3 & 1 & 3 \\ 3 & 2 & 2 \end{bmatrix}$$

We can further expand these 3×3 determinants along rows or columns. For instance, the first can be expanded along the last column, which has only a single nonzero entry, and the second can be expanded along the first row:

$$-4 \left(2 \det \begin{bmatrix} 1 & 3 \\ 2 & 2 \end{bmatrix} + 0 + 0 \right) + 0 - 0 + 3 \left(1 \det \begin{bmatrix} 1 & 3 \\ 2 & 2 \end{bmatrix} - 0 + 1 \det \begin{bmatrix} 3 & 1 \\ 3 & 2 \end{bmatrix} \right)$$

This comes out to

$$-4(2(2-6)) + 3((2-6) + (6-3)) = 32 - 3 = 29$$

Why do these formulas compute the determinant? This can be explained in terms of the definition of the determinant as a sum over permutations. Consider the case of the expansion across the first row. Each permutation selects one element from the first row. We can group permutations according to which element they select. After we have selected an element of the first row, say a_{1j} , we select other entries of the matrix whose row and column are different; these correspond to terms in the determinant of A_{1j} . [Draw picture]

4.5 Determinants and Row Operations

For us, much of the meaning and usefulness of the determinant results from how it is related to row operations. The important properties are recorded below:

1. If B is obtained by swapping two rows of A , then $\det(B) = -\det(A)$.
2. If B is obtained by multiplying a row of A by a number k , then $\det(B) = k \det(A)$.

3. If A_1 and A_2 share all rows except the i th and B is obtained by replacing this i th row with the sum of that of A_1 and A_2 , then

$$\det(B) = \det(A_1) + \det(A_2)$$

4. If A has a repeated row, then $\det(A) = 0$.

Each of these properties is a consequence of the definition of determinant as a sum over permutations. As an example, for (1), if we swap two rows, this changes the sign on every term in the determinant. (4) follows from (1), as swapping the repeated rows yields $\det(A) = -\det(A)$.

A useful consequence of (2), (3), and (4) is that the determinant does not change if we add a multiple of one row to another. If B is obtained from A by adding $k \cdot r_i$ to r_j , then $\det(B) = \det(A) + k \cdot \det(C)$, where in C , the row r_i is repeated. Thus $\det(C) = 0$, and so $\det(B) = \det(A)$.

Remark. All the properties listed above are true for operations on columns as well as rows.

With these observations, we can justify the most important property of the determinant.

Proposition. *An $n \times n$ matrix A is invertible if and only if $\det(A) \neq 0$.*

Proof Sketch. If A is invertible, then there is a sequence of row operations bringing it to the identity matrix. Each row operation either multiplies the determinant by -1 , if it swaps rows, by a nonzero k , if it scales a row, or does not change it at all, if it adds one row to another. Because the determinant of the identity matrix is $1 \neq 0$, the determinant of the original matrix had to be nonzero as well.

If A is not invertible, then because it is square, there must be a zero row in its reduced row echelon form R . Each permutation selects an entry from this row, so all the products are 0 and so $\det(R) = 0$. As explained above, this means that $\det(A)$ had to have been 0 as well. $\square$

If we keep track of how the determinant changes as we apply row operations, we can use this to compute it, by reducing the matrix to the identity matrix where we know it is 1. In fact, we don't even need to go all the way to the identity matrix; once we get it to an upper triangular matrix, we can compute the determinant by taking the product of the diagonal entries. This is usually the most efficient way to compute determinants of large matrices.

Example 10. Let's compute the determinant

$$\det \begin{bmatrix} 0 & 7 & 5 & 3 \\ 2 & 2 & 4 & 2 \\ 1 & 1 & 2 & -1 \\ 1 & 1 & 1 & 2 \end{bmatrix}$$

using row operations. I will do this in not necessarily the most efficient way, in order to illustrate the method.

$$\begin{aligned} \det \begin{bmatrix} 0 & 7 & 5 & 3 \\ 2 & 2 & 4 & 2 \\ 1 & 1 & 2 & -1 \\ 1 & 1 & 1 & 2 \end{bmatrix} &= -\det \begin{bmatrix} 2 & 2 & 4 & 2 \\ 0 & 7 & 5 & 3 \\ 1 & 1 & 2 & -1 \\ 1 & 1 & 1 & 2 \end{bmatrix} = -2 \det \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & 7 & 5 & 3 \\ 1 & 1 & 2 & -1 \\ 1 & 1 & 1 & 2 \end{bmatrix} = -2 \det \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & 7 & 5 & 3 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & -1 & 1 \end{bmatrix} \\ &= 2 \det \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & 7 & 5 & 3 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & -2 \end{bmatrix} = 2(1 \cdot 7 \cdot (-1) \cdot (-2)) = 28 \end{aligned}$$